

I2 Numerical solutions A level only

Name: _____

Class: _____

Date: _____

Time: **189 minutes**

Marks: **157 marks**

Comments:

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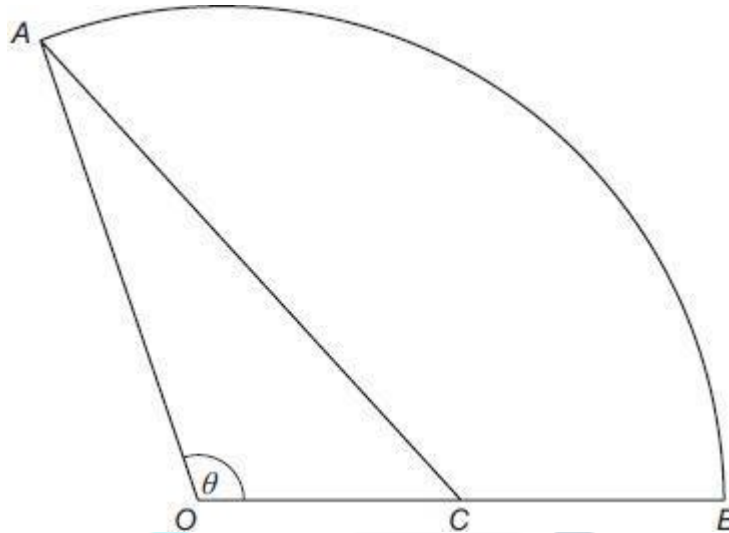
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Q1.

The diagram shows a sector of a circle OAB .

C is the midpoint of OB .

Angle AOB is θ radians.



- (a) Given that the area of the triangle OAC is equal to one quarter of the area of the sector OAB , show that $\theta = 2 \sin \theta$ (4)
- (b) Use the Newton-Raphson method with $\theta_1 = \pi$, to find θ_3 as an approximation for θ . Give your answer correct to five decimal places. (3)
- (c) Given that $\theta = 1.89549$ to five decimal places, find an estimate for the percentage error in the approximation found in part (b). (1)

(Total 8 marks)

Q2.

The daily world production of oil can be modelled using

$$V = 10 + 100\left(\frac{t}{30}\right)^3 - 50\left(\frac{t}{30}\right)^4$$

where V is volume of oil in millions of barrels, and t is time in years since 1 January 1980.

- (a) (i) The model is used to predict the time, T , when oil production will fall to zero.

Show that T satisfies the equation

$$T = \sqrt[3]{60T^2 + \frac{162\,000}{T}}$$

(3)

- (ii) Use the iterative formula $T_{n+1} = \sqrt[3]{60T_n^2 + \frac{162\,000}{T_n}}$, with $T_0 = 38$, to find the values of T_1 , T_2 , and T_3 , giving your answers to three decimal places.

(2)

- (iii) Explain the relevance of using $T_0 = 38$

(1)

- (b) From 1 January 1980 the daily use of oil by one technologically developing country can be modelled as

$$V = 4.5 \times 1.063^t$$

Use the models to show that the country's use of oil and the world production of oil will be equal during the year 2029.

(4)

(Total 10 marks)

Q3.

The equation $x^3 - 3x + 1 = 0$ has three real roots.

- (a) Show that one of the roots lies between -2 and -1
- (b) Taking $x_1 = -2$ as the first approximation to one of the roots, use the Newton-Raphson method to find x_2 , the second approximation.
- (c) Explain why the Newton-Raphson method fails in the case when the first approximation is $x_1 = -1$

(2)

(3)

(1)

(Total 6 marks)

Q4.

- (a) Show that the equation $x^3 - 6x + 1 = 0$ has a root α , where $2 < \alpha < 3$.
- (b) Show that the equation $x^3 - 6x + 1 = 0$ can be rearranged into the form

(2)

$$x^2 = 6 - \frac{1}{x}$$

(1)

- (c) Use the recurrence relation $x_{n+1} = \sqrt{6 - \frac{1}{x_n}}$, with $x_1 = 2.5$, to find the value of x_3 , giving your answer to four significant figures.

(2)

(Total 5 marks)

Q5.

- (a) The equation $e^{-x} - 2 + \sqrt{x} = 0$ has a single root, α .

Show that α lies between 3 and 4.

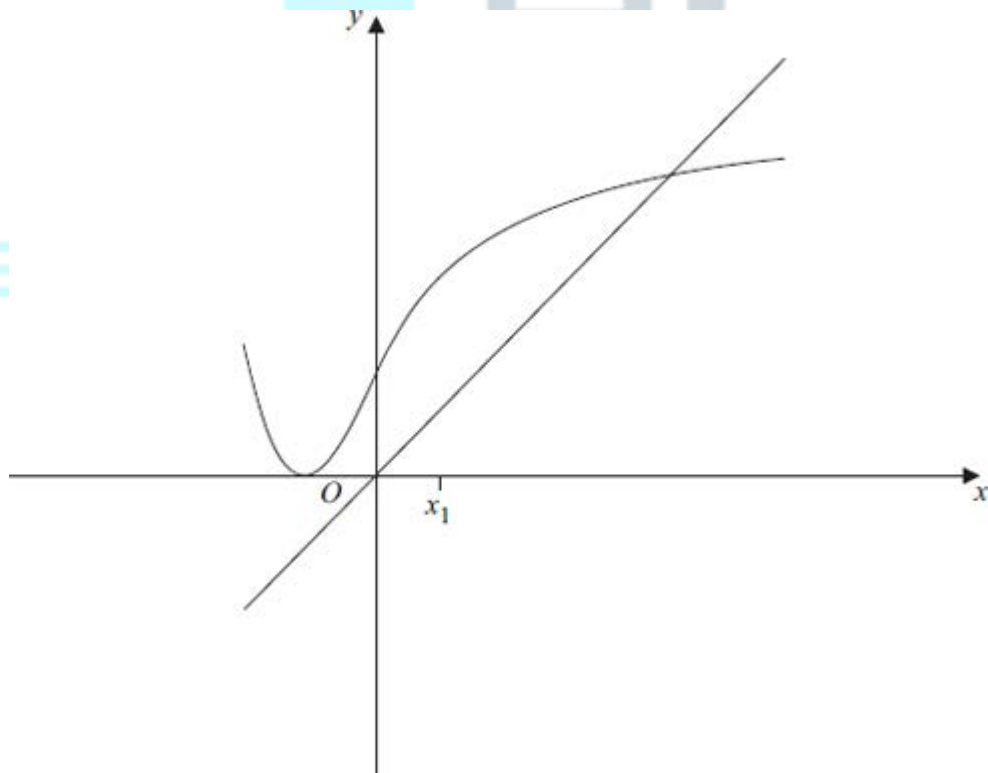
(2)

- (b) Use the recurrence relation $x_{n+1} = (2 - e^{-x_n})^2$, with $x_1 = 3.5$, to find x_2 and x_3 , giving your answers to three decimal places.

(2)

- (c) The diagram below shows parts of the graphs of $y = (2 - e^{-x})^2$ and $y = x$, and a position of x_1 .

On the diagram, draw a staircase or cobweb diagram to show how convergence takes place, indicating the positions of x_2 and x_3 on the x -axis.



(2)

(Total 6 marks)

Q6.

The equation

$$x^3 - x^2 + 4x - 900 = 0$$

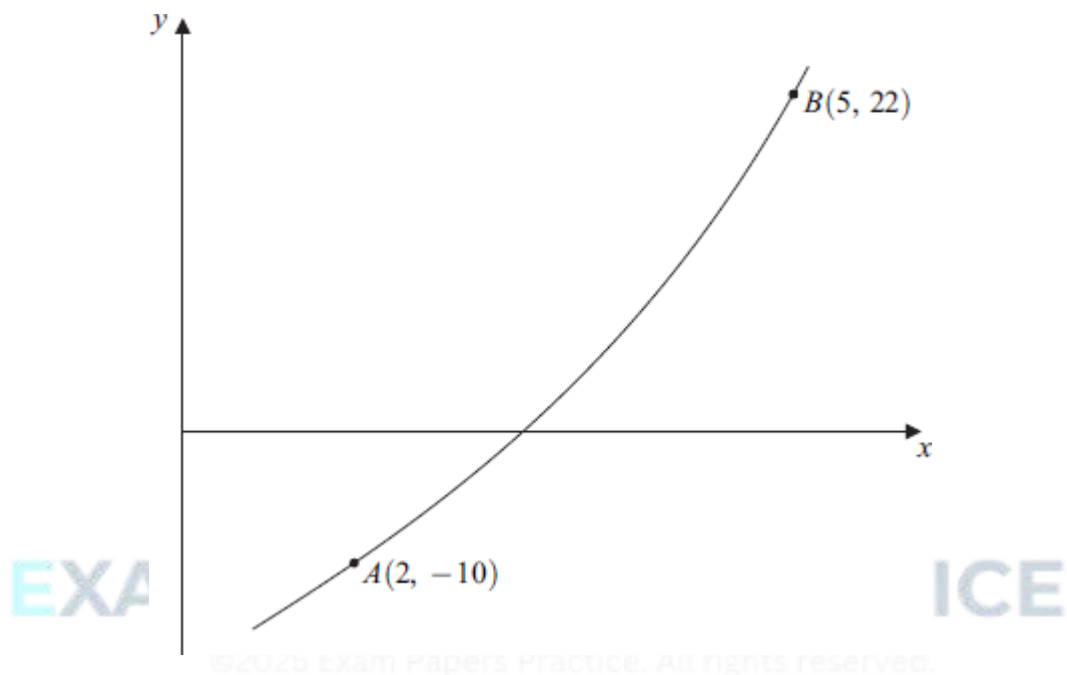
has exactly one real root, α .

Taking $x_1 = 10$ as a first approximation to α , use the Newton–Raphson method to find a second approximation, x_2 , to α . Give your answer to four significant figures.

(Total 3 marks)

Q7.

The diagram below (not to scale) shows a part of a curve $y = f(x)$ which passes through the points $A(2, -10)$ and $B(5, 22)$.



- (a) On the same diagram, draw a line which illustrates the Newton–Raphson method for solving the equation $f(x) = 0$, with initial value $x_1 = 2$. The point of intersection of this line with the x -axis should be labelled Q .

(1)

- (b) Given that the gradient of the curve at A is 8, calculate the x -coordinate of Q . Give your answer as an exact decimal.

(2)

(Total 3 marks)

Q8.

For $0 < x \leq 2$, the curves with equations $y = 4 \ln x$ and $y = \sqrt{x}$ intersect at a single point where $x = \alpha$.

- (a) Show that α lies between 0.5 and 1.5.

(2)

- (b) Show that the equation $4 \ln x = \sqrt{x}$ can be rearranged into the form

$$x = e^{\left(\frac{\sqrt{x}}{4}\right)}$$

(1)

- (c) Use the iterative formula

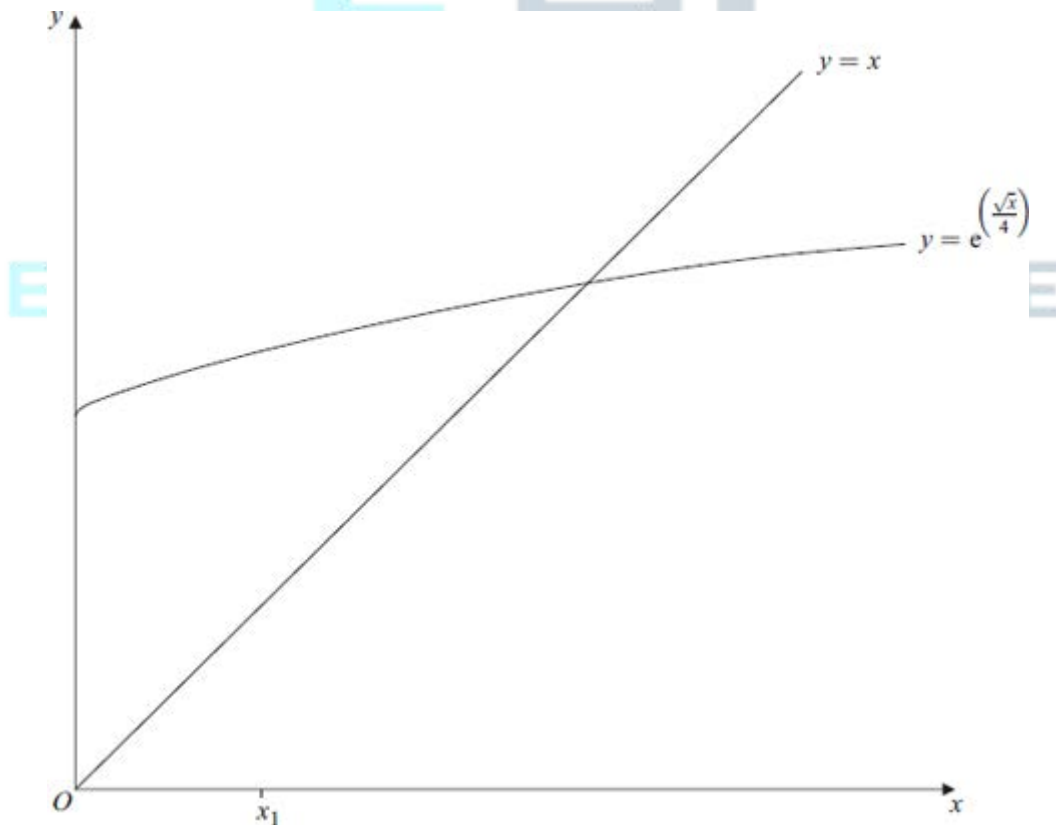
$$x_{n+1} = e^{\left(\frac{\sqrt{x_n}}{4}\right)}$$

with $x_1 = 0.5$ to find the values of x_2 and x_3 , giving your answers to three decimal places.

(2)

- (d) The figure below shows a sketch of parts of the graphs of $y = e^{\left(\frac{\sqrt{x}}{4}\right)}$ and $y = x$, and the position of x_1 .

Draw a cobweb or staircase diagram on the figure below to show how convergence takes place, indicating the positions of x_2 and x_3 on the x -axis.



(2)

(Total 7 marks)

Q9.

The equation

$$24x^3 + 36x^2 + 18x - 5 = 0$$

has one real root, α .

- (a) Show that α lies in the interval $0.1 < x < 0.2$.

(2)

- (b) Taking $x_1 = 0.2$ as a first approximation to α , use the Newton-Raphson method to find a second approximation, x_2 , to α . Give your answer to four decimal places.

(4)

(Total 6 marks)

Q10.

A curve is defined by the equation $y = (x^2 - 4) \ln(x + 2)$ for $x \geq 3$.

The curve intersects the line $y = 15$ at a single point, where $x = \alpha$.

- (a) Show that α lies between 3.5 and 3.6.

(2)

- (b) Show that the equation $(x^2 - 4) \ln(x + 2) = 15$ can be arranged into the form

$$x = \pm \sqrt{4 + \frac{15}{\ln(x+2)}}$$

(2)

- (c) Use the iteration

$$x_{n+1} = \sqrt{4 + \frac{15}{\ln(x_n + 2)}}$$

with $x_1 = 3.5$ to find the values of x_2 and x_3 , giving your answers to three decimal places.

(2)

(Total 6 marks)

Q11.

The curve $y = \cos^{-1}(2x - 1)$ intersects the curve $y = e^x$ at a single point where $x = \alpha$.

- (a) Show that α lies between 0.4 and 0.5.

(2)

- (b) Show that the equation $\cos^{-1}(2x - 1) = e^x$ can be written as $x = \frac{1}{2} + \frac{1}{2} \cos(e^x)$. (1)

- (c) Use the iteration $x_{n+1} = \frac{1}{2} + \frac{1}{2} \cos(e^{x_n})$ with $x_1 = 0.4$ to find the values of x^2 and x^3 giving your answers to three decimal places. (2)

(Total 5 marks)

Q12.

- (a) (i) Sketch the graph of $y = \sin^{-1} x$, where y is in radians. State the coordinates of the end points of the graph. (3)
- (ii) By drawing a suitable straight line on your sketch, show that the equation

$$\sin^{-1} x = \frac{1}{4} x + 1$$

has only one solution. (2)

- (b) The root of the equation $\sin^{-1} x = \frac{1}{4} x + 1$ is α . Show that $0.5 < \alpha < 1$. (2)

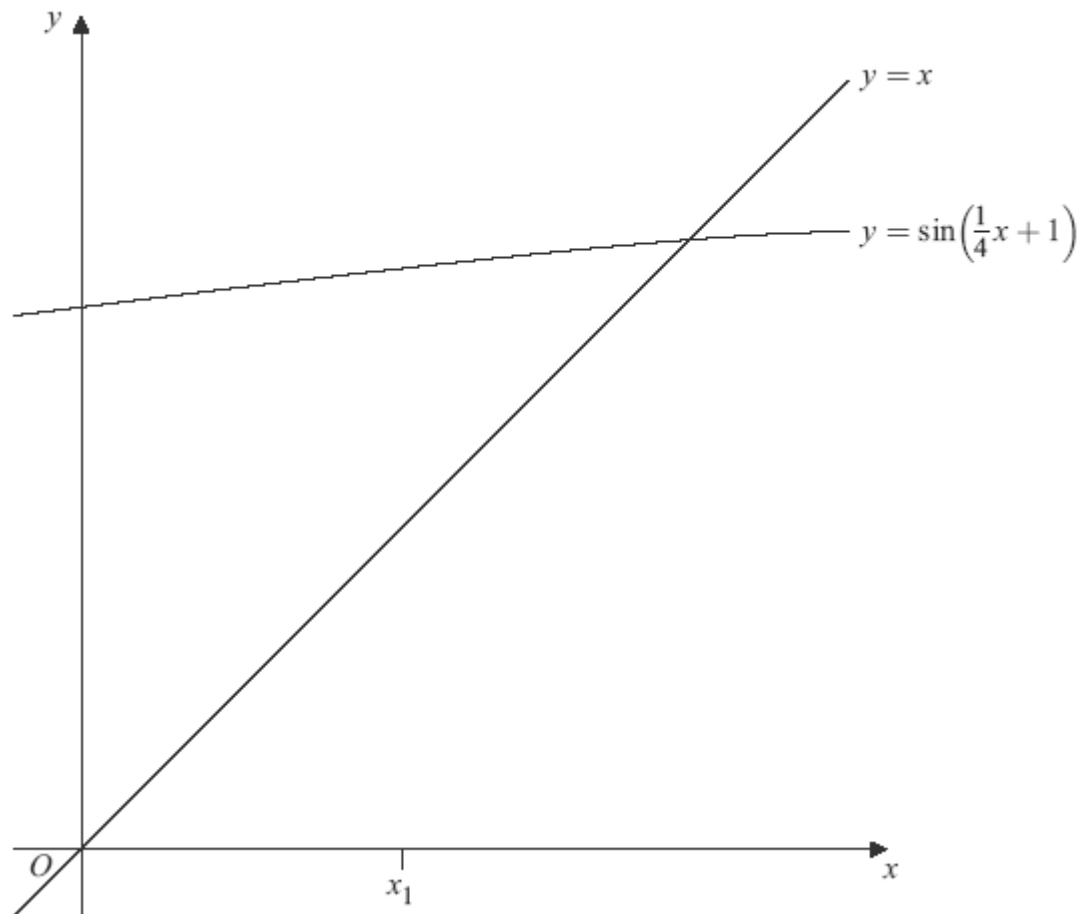
- (c) The equation $\sin^{-1} x = \frac{1}{4} x + 1$ can be rewritten as $x = \sin\left(\frac{1}{4} x + 1\right)$. (2)

- (i) Use the iteration $x_{n+1} = \sin\left(\frac{1}{4} x_n + 1\right)$ with $x_1 = 0.5$ to find the values of x_2 and x_3 , giving your answers to three decimal places. (2)

- (ii) The sketch on the figure below shows parts of the graphs of

$$y = \sin\left(\frac{1}{4} x + 1\right) \quad \text{and} \quad y = x, \quad \text{and the position of } x_1.$$

On the figure below, draw a cobweb or staircase diagram to show how convergence takes place, indicating the positions of x_2 and x_3 on the x -axis.



(2)
(Total 11 marks)

Q13.

The curve $y = 3^x$ intersects the curve $y = 10 - x^3$ at the point where $x = \alpha$.

- (a) Show that α lies between 1 and 2.

(2)

- (b) (i) Show that the equation $3^x = 10 - x^3$ can be rearranged into the form

$$x = \sqrt[3]{10 - 3^x}.$$

(1)

- (ii) Use the iteration $x_{n+1} = \sqrt[3]{10 - 3^{x_n}}$ with $x_1 = 1$ to find the values of x_2 and x_3 giving your answers to three decimal places.

(2)
(Total 5 marks)

Q14.

The curve with equation $y = x^3 + 5x - 4$ intersects the x -axis at the point A , where $x = \alpha$.

- (a) Show that α lies between 0.5 and 1.

(2)

- (b) Show that the equation $x^3 + 5x - 4 = 0$ can be rearranged into the form

$$x = \frac{1}{5}(4 - x^3)$$

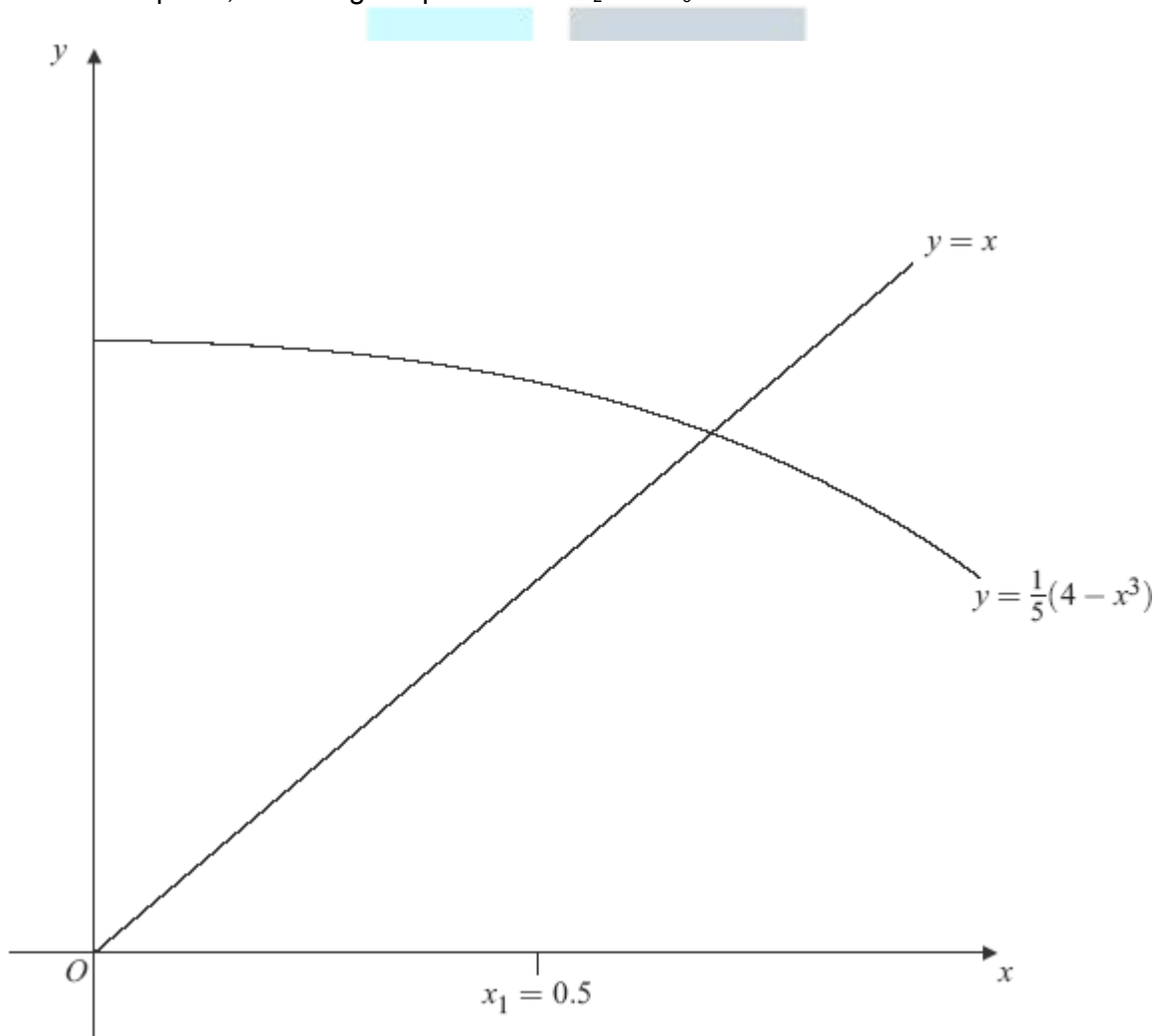
(1)

- (c) Use the iteration $x_{n+1} = \frac{1}{5}(4 - x_n^3)$ with $x_1 = 0.5$ to find x_3 , giving your answer to three decimal places.

(2)

- (d) The sketch on the figure below shows parts of the graphs of $y = \frac{1}{5}(4 - x^3)$ and $y = x$, and the position of x_1 .

On the figure below, draw a cobweb or staircase diagram to show how convergence takes place, indicating the positions of x_2 and x_3 on the x -axis.



(2)
(Total 7 marks)

Q15.

- (a) The curve with equation

$$y = \frac{\cos x}{2x+1}, \quad x > -\frac{1}{2}$$

intersects the line $y = \frac{1}{2}$ at the point where $x = \alpha$.

- (i) Show that α lies between 0 and $\frac{\pi}{2}$.

(2)

- (ii) Show that the equation $\frac{\cos x}{2x+1} = \frac{1}{2}$ can be rearranged into the form

$$x = \cos x - \frac{1}{2}$$

(1)

- (iii) Use the iteration $x_{n+1} = \cos x_n - \frac{1}{2}$ with $x_1 = 0$ to find x_3 , giving your answer to three decimal places.

(2)

- (b) (i) Given that $y = \frac{\cos x}{2x+1}$, use the quotient rule to find an expression for $\frac{dy}{dx}$.

(3)

- (ii) Hence find the gradient of the normal to the curve $y = \frac{\cos x}{2x+1}$ at the point on the curve where $x = 0$.

(2)

(Total 10 marks)

Q16.

The equation

$$x + (1 + 3x)^{\frac{1}{4}} = 0$$

has a single root, α .

- (a) Show that α lies between -0.33 and -0.32 .

(2)

- (b) Show that the equation $x + (1 + 3x)^{\frac{1}{4}} = 0$ can be rearranged into the form

$$x = \frac{1}{3}(x^4 - 1)$$

(2)

- (c) Use the iteration $x_{n+1} = \frac{(x_n^4 - 1)}{3}$ with $x_1 = -0.3$ to find x_4 , giving your answer to three significant figures.

(3)

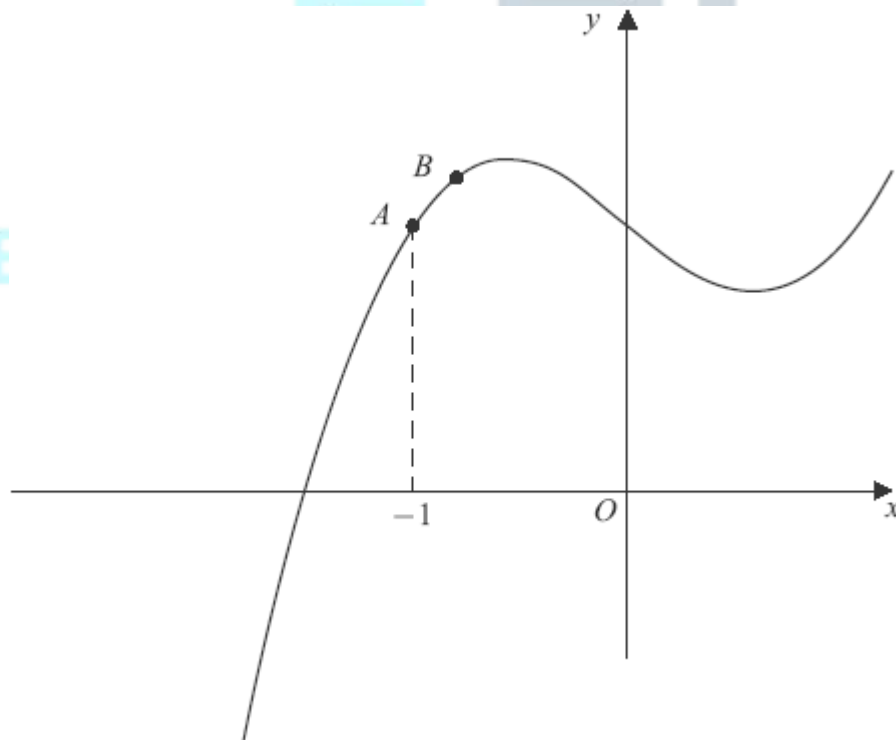
(Total 7 marks)

Q17.

The diagram shows the curve

$$y = x^3 - x + 1$$

The points A and B on the curve have x -coordinates -1 and $-1 + h$ respectively.



- (a) (i) Show that the y -coordinate of the point B is

$$1 + 2h - 3h^2 + h^3$$

(3)

- (ii) Find the gradient of the chord AB in the form

$$p + qh + rh^2$$

where p , q and r are integers.

(3)

- (iii) Explain how your answer to part (a)(ii) can be used to find the gradient of the tangent to the curve at A . State the value of this gradient.

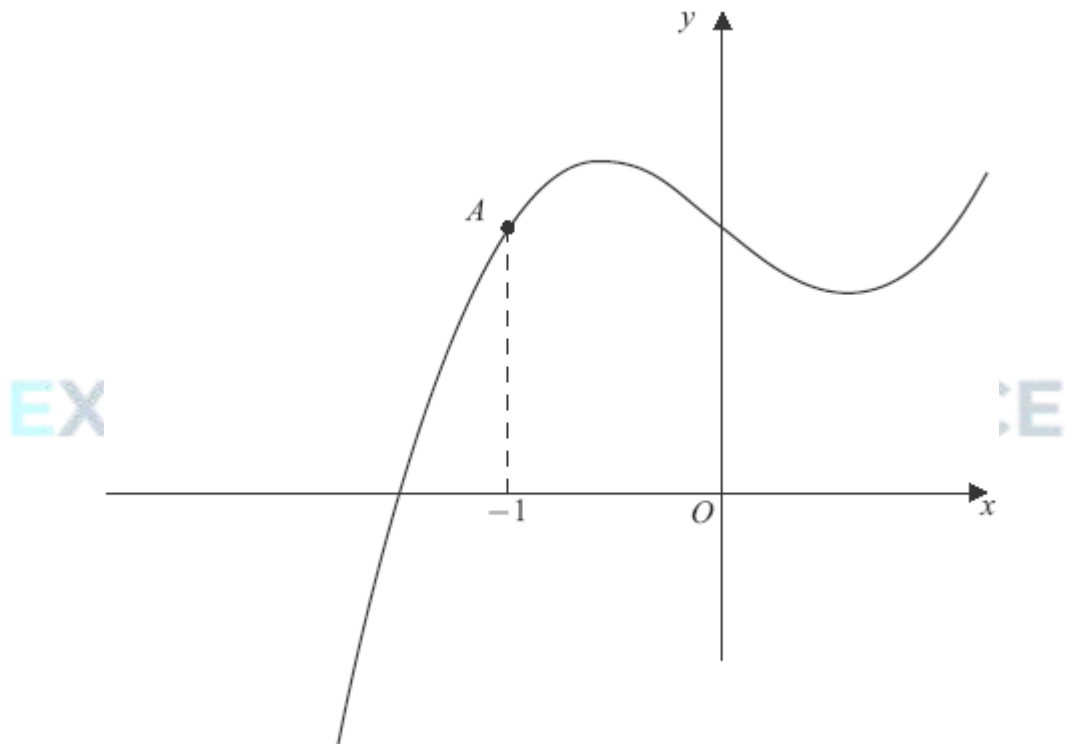
(2)

- (b) The equation $x^3 - x + 1 = 0$ has one real root, α .

- (i) Taking $x_1 = -1$ as a first approximation to α , use the Newton-Raphson method to find a second approximation, x_2 , to α .

(2)

- (ii) On the figure below, draw a straight line to illustrate the Newton-Raphson method as used in part (b)(i). Show the points $(x_2, 0)$ and $(\alpha, 0)$ on your diagram.

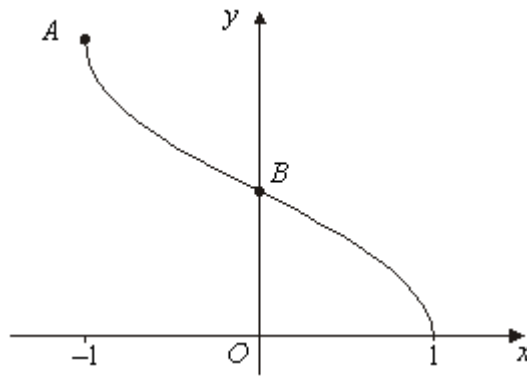


(2)

(Total 12 marks)

Q18.

The diagram shows the curve $y = \cos^{-1}x$ for $-1 \leq x \leq 1$



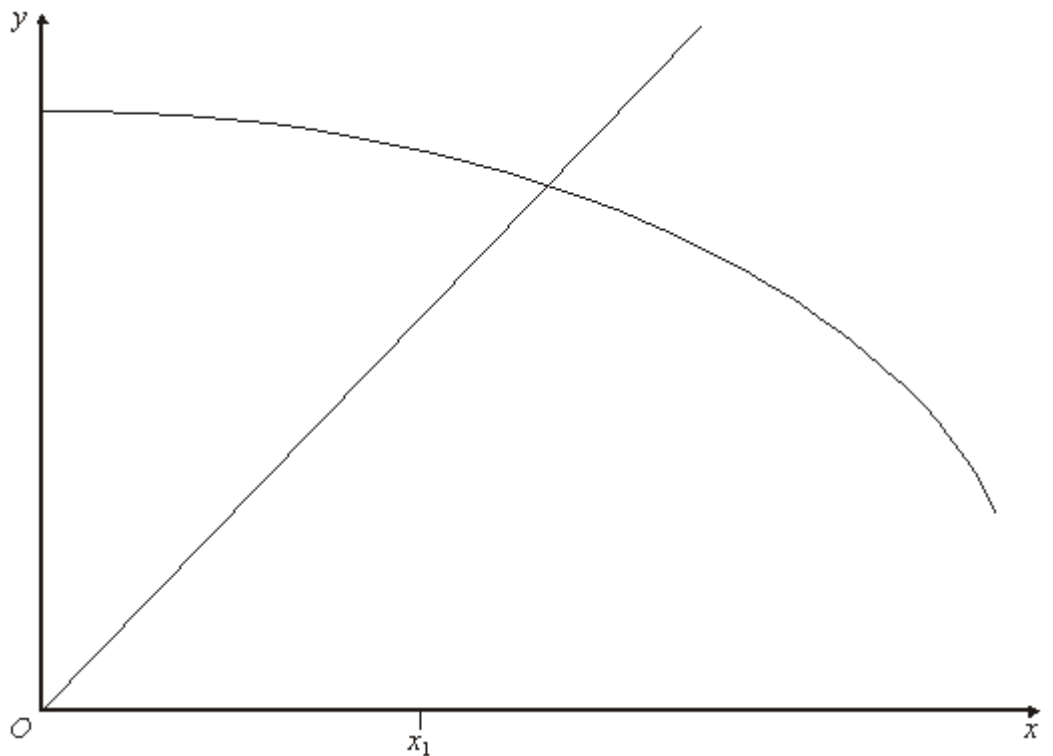
- (a) Write down the exact coordinates of the points A and B . (2)
- (b) The equation $\cos^{-1} x = 3x + 1$ has only one root. Given that the root of this equation is α , show that $0.1 \leq \alpha \leq 0.2$. (2)
- (c) Use the iteration $x_{n+1} = \frac{1}{3}(\cos^{-1} x_n - 1)$ with $x_1 = 0.1$ to find the values of x_2 , x_3 and x_4 , giving your answers to three decimal places. (3)
- (Total 7 marks)

Q19.

The curve $y = x^3 + 4x - 3$ intersects the x -axis at the point A where $x = \alpha$.

- (a) Show that α lies between 0.5 and 1.0. (2)
- (b) Show that the equation $x^3 + 4x - 3 = 0$ can be rearranged into the form $x = \frac{3 - x^3}{4}$. (1)
- (c) (i) Use the iteration $x_{n+1} = \frac{3 - x_n^3}{4}$ with $x_1 = 0.5$ to find x_3 , giving your answer to two decimal places. (3)
- (ii) The sketch on the figure below shows parts of the graphs of $y = \frac{3 - x^3}{4}$ and $y = x$, and the position of x_1 .

On the figure below, draw a cobweb or staircase diagram to show how convergence takes place, indicating the positions of x_2 and x_3 on the x -axis.



(3)
(Total 9 marks)

Q20.

The curve $y = x^3 - x - 7$ intersects the x-axis at the point where $x = a$.

- (a) Show that a lies between 2.0 and 2.1.

(2)

- (b) Show that the equation $x^3 - x - 7 = 0$ can be rearranged in the form $x = \sqrt[3]{x+7}$.

(1)

- (c) Use the iteration $x_{n+1} = \sqrt[3]{x_n + 7}$ with $x_1 = 2$ to find the values of x_2 , x_3 and x_4 , giving your answers to three significant figures.

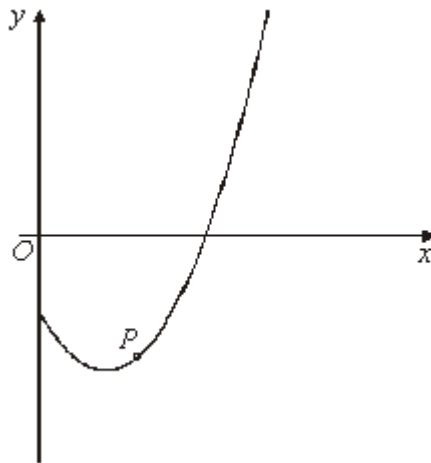
(3)
(Total 6 marks)

Q21.

The diagram shows a part of the graph of $y = f(x)$, where

$$f(x) = x^3 - 2x - 1$$

The point P has coordinates $(1, -2)$.



- (a) Taking $x_1 = 1$ as a first approximation to a root of the equation $f(x) = 0$, use the Newton-Raphson method to find a second approximation, x_2 , to the root. (3)
- (b) On the figure above, draw a straight line to illustrate the Newton-Raphson method as used in part (a).
Mark x_1 and x_2 on the figure. (2)
- (c) By considering $f(2)$, show that the second approximation found in part (a) is not as good as the first approximation. (2)
- (d) Taking $x_1 = 1.6$ as a first approximation to the root, use the Newton-Raphson method to find a second approximation to the root. Give your answer to three decimal places. (2)
- (Total 9 marks)

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Q22.

- (a) Sketch the graph of $y = \tan^{-1}x$. (2)
- (b) (i) By drawing a suitable straight line on your sketch, show that the equation $\tan^{-1}x = 2x - 1$ has only one root. (2)
- (ii) Given that the root of this equation is α , show that $0.8 < \alpha < 0.9$. (2)
- (c) Use the iteration $x_{n+1} = \frac{1}{2}(\tan^{-1}x_n + 1)$ with $x_1 = 0.8$ to find the value of x_3 , giving your answer to two significant figures. (3)

(Total 9 marks)



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